```
\\ Global Definition
IMAGE_SIZE = batch * channels * height * width
FEATURE_SIZE = num_levels * batch * feature_dim * height * width
QUERY_SIZE = batch * num_queries * feature_dim
REF_POINT_SIZE = batch * num_queries * num_levels * 2

\\ Initialize Ref Points and Clustering for 20% Queries
Host::queries, ref_points = initialize_queries()
Host::offsets = Mat_Mul(queries, weights)
Host::centers = clustering(ref_points + offsets)

\\ Memory Allocation
Host::MemAlloc(image, IMAGE_SIZE)
Host::MemAlloc(ms_features, FEATURE_SIZE, centers)
Host::MemAlloc(queries & probability, QUERY_SIZE)
Host::MemAlloc(ref_points, REF_POINT_SIZE)

\\ Position Encoding and Calculate Probability
Host::features = backbone(image)
Host::ms_features = position_encoding(features)
Host::probability = softmax(Mat_Mul(queries, attn_weights))

\\ NMP Kernels for MSDAttn
NMP::MSDAttn(ms_features, probability, ref_points, offsets){
    memory = transformer_encoder(ms_features)
    for decoder in decoder_layers:
        sampled_features = bilinear_interpolation(memory, ref_points, offsets)
        output = weighted_sum(sampled_features, probability)
}

Host::pred_bboxes, pred_classes = predict(output)
Host::detections = filter_detections(pred_bboxes, pred_classes)
```